Fiscal Information in Monetary Surprises

Markeljan Blacëri*

October 2026

Abstract

Are informationally robust monetary policy instruments contaminated by the Fed's view of fiscal policy? Miranda-Agrippino and Ricco (2021) purge high-frequency surprises of the Fed's macroeconomic forecasts, but note that the instrument would remain contaminated if the Fed also responded to fiscal variables. I test this by adding the Greenbook forecasts of the federal surplus and of the high-employment surplus, extended to 2020, to their information set. The fiscal forecasts do not predict the surprises, the instrument is essentially unchanged, and the responses to a monetary policy shock in a proxy VAR differ little. The Fed's fiscal outlook leaves no detectable trace in the surprises, and the informationally robust instrument needs no fiscal correction.

Keywords: monetary policy shocks, central bank information, fiscal forecasts
JEL: E52, E58, E62, C32, C36

*Department of Economics and Management, University of Padova, Via del Santo 33, 35123 Padova, Italy. Email: markeljan.blaceri@phd.unipd.it.
The updated instruments can be found at https://markeljanblaceri.github.io/data.

1 Introduction

High-frequency changes in interest rate futures around FOMC announcements are the standard external instrument for monetary policy shocks (Kuttner, 2001; Gürkaynak et al., 2005; Gertler and Karadi, 2015). The identifying assumption is that, within a narrow window around the announcement, these price changes reflect only the unexpected component of the policy decision. A large literature has shown that this assumption fails in two ways. First, FOMC announcements also convey the Fed's assessment of the economic outlook, so market surprises combine policy shocks with central bank information effects (Romer and Romer, 2000; Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021). Second, the surprises are predictable from information available before the announcement (Miranda-Agrippino and Ricco, 2021; Bauer and Swanson, 2023).

Miranda-Agrippino and Ricco (2021) (henceforth MAR) address both problems. To remove the component of the surprises that reflects the Fed's private information, they project surprises in the fourth federal funds futures contract on the Greenbook forecasts, and revisions to those forecasts, for output growth, inflation and unemployment. Because market participants absorb information only gradually, the residuals from this projection are autocorrelated, so MAR also remove their autoregressive component. The resulting instrument is informationally robust, meaning that it is orthogonal both to the Fed staff's macroeconomic forecasts and to its own past values.

The projection, however, conditions only on the Greenbook's macroeconomic forecasts. As MAR themselves note, if the central bank sets the policy rate conditioning on other indicators, such as financial and fiscal variables, the projection residuals will also be endogenous to these variables (Miranda-Agrippino and Ricco, 2021, footnote 16). Whether this is the case for the fiscal variables has not been tested. The reason is a practical one. The Greenbook data set distributed by the Philadelphia Fed, on which the projection is built, does not contain the staff's forecasts of the federal budget.

The concern is a plausible one. Croushore and van Norden (2018) collect the staff's fiscal forecasts from the Greenbooks and show that the forecasts of the federal surplus and of the high-employment surplus help explain the FOMC's policy decisions.[1] If the Fed responds to its fiscal outlook, and if the FOMC announcements reveal that outlook

[1] They add the fiscal forecasts to the regression of Romer and Romer (2004), which relates the change in the intended federal funds rate at each FOMC meeting to the Greenbook forecasts of output growth, inflation and unemployment.

to markets, the surprises would contain a fiscal information component alongside the macroeconomic one, and the MAR instrument would inherit it. Even a small contamination of this kind can be costly. Miranda-Agrippino and Ricco (2023) show that a proxy only weakly correlated with a non-target shock can still produce a large bias in the estimated impulse responses, and that the bias grows with the share of the variance of the variables of interest explained by the non-target shock relative to the target one. In a monetary VAR, fiscal shocks are a natural candidate for such a non-target shock.

This note tests whether the MAR instrument contains such a fiscal component. I extend the Greenbook fiscal forecasts of Croushore and van Norden (2018) from December 2010 to December 2020. For 2011-2012, I use the federal sector accounts provided with their replication files. From 2013 onward, I read the same staff table, *"Staff Projections of Federal Sector Accounts and Related Items"*, from each Tealbook.[2] I then add the forecasts of the surplus and of the high-employment surplus, as ratios to GDP, to the information set of the MAR projection, in levels for the previous, current and next three quarters and in revisions for the previous, current and next two quarters, exactly as the macroeconomic forecasts enter. The surprises are those of Jarociński and Karadi (2020), which cover every FOMC announcement from 1990 to 2020.

The answer is no. Conditional on the Greenbook macroeconomic forecasts, the fiscal forecasts have no predictive power for the monetary surprises. The joint tests that all fiscal coefficients are zero are far from rejecting the null in every specification, whether the surplus, the high-employment surplus, or both are added, and whether the sample ends before the zero lower bound, before the redesign of the Tealbook fiscal table in 2017, or in 2020. As a result, including the fiscal forecasts leaves the monetary policy instrument essentially unchanged, and the versions built with and without them are almost perfectly correlated. In a proxy VAR, the instruments built with and without the fiscal forecasts have similar strength and deliver similar responses to a monetary policy shock.

The result does not say that the Fed ignores fiscal policy. Croushore and van Nor-

[2] All credit for the construction of the fiscal series goes to Croushore and van Norden (2018); this note only extends their data. Their replication files are available at https://svannorden.org/code-and-data/. The Tealbooks are published by the Board of Governors of the Federal Reserve System at https://www.federalreserve.gov/monetarypolicy/fomc_historical.htm. On every Greenbook from 2002 to 2012 for which the sources overlap, the extended series coincide exactly with theirs. From September 2017, the Tealbook reports the unified federal budget by fiscal year and a cyclically adjusted rather than a high-employment surplus; Section 2 describes how the two tables are linked.

den (2018) study the intended policy rate, that is, the reaction function, whereas the MAR projection is run on market surprises. A systematic response of the FOMC to the fiscal outlook does not generate predictable surprises if markets anticipate it, nor a fiscal information effect if the staff's fiscal forecasts tell markets nothing they did not already know about the budget. The null is therefore informative about the instrument rather than about the policy rule. The informationally robust instrument of MAR needs no fiscal correction, and the concern raised in their footnote 16 does not apply to the fiscal side of the Greenbook.

The note contributes to the literature on information effects in monetary surprises (Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021; Bauer and Swanson, 2023) and to the work on the Fed's attention to fiscal policy (Croushore and van Norden, 2018, 2019).

The rest of this note is organised as follows. Section 2 describes the monetary surprises, the Greenbook macroeconomic and fiscal forecasts, and the construction of the informationally robust instrument. Section 3 tests whether the fiscal forecasts predict the surprises. Section 4 compares the responses to a monetary policy shock identified with the instruments built with and without the fiscal forecasts. Section 5 concludes.

2 Fiscal information and the informationally-robust IV

This section describes the data and the procedure of Miranda-Agrippino and Ricco (2021), which I apply to the high-frequency monetary policy surprises in the replication files of Jarociński and Karadi (2020). Following their method, the surprises are purged of the component predictable from the Fed's Greenbook forecasts and forecast revisions, and of their own autoregressive component. The only change is to the information set of the first step, to which I add the Greenbook forecasts of the federal surplus and of the high-employment surplus, collected by Croushore and van Norden (2018) and extended here to 2020. I then test whether these forecasts predict the surprises.

2.1 Monetary policy surprises

Following Miranda-Agrippino and Ricco (2021), the surprise measure is the change in the price of the fourth federal funds futures contract (FF4) in a narrow window around

each FOMC announcement. I take the surprises from the replication data of Jarociński and Karadi (2020), which include both scheduled and unscheduled announcements.[3] Surprises are measured over a 30-minute window that opens 10 minutes before the announcement. I use observations from 1990 to 2020. Six days in the sample contain two announcements;[4] on these days I sum the two surprises and treat them as a single event.

2.2 Greenbook forecasts

Before each scheduled FOMC meeting, the staff of the Board of Governors prepare a detailed forecast of the U.S. economy, published in the Greenbook.[5] The forecasts are made public with a five-year lag. The macroeconomic forecasts are taken from the Tealbook (formerly Greenbook) Data Sets of the Federal Reserve Bank of Philadelphia. As in Miranda-Agrippino and Ricco (2021), I use the forecasts of real GDP growth, GDP-deflator inflation and the unemployment rate. The forecasts are quarterly and refer to the quarter before the Greenbook, the current quarter and the following quarters.

Alongside the macroeconomic forecasts, the staff also prepare projections of the federal budget. These projections are not part of the Philadelphia Fed data sets. Croushore and van Norden (2018) collect them from the Greenbooks from December 1966 to December 2010. I extend their series to December 2020, using the federal sector accounts provided with their replication files for 2011–2012 and, from 2013 onward, the same staff table read from each Tealbook A.[6]

From the fiscal projections, I use two variables, the forecast of the federal surplus and the forecast of the high-employment surplus, that is, the surplus that would prevail with the economy at its potential. Until July 2017, both refer to the federal sector of the national income and product accounts and are reported by calendar quarter in

[3]The surprises are available at `https://github.com/marekjarocinski/jkshocks_update_fed`.

[4]Five of these days, all in 1991, featured two open market operations; on 3 May 2005, the FOMC statement was followed by a correction.

[5]In June 2010 the Greenbook and the Bluebook were merged into a single report, the Tealbook. The staff forecasts previously published in the Greenbook are contained in Tealbook A ("Economic and Financial Conditions: Current Situation and Outlook"), so the forecast series is continuous across the change. Following Miranda-Agrippino and Ricco (2021), I refer to the forecasts as the Greenbook throughout.

[6]On every Greenbook from 2002 to 2012 for which the sources overlap, the values I extract coincide with theirs.

billions of dollars at a seasonally adjusted annual rate. As in Croushore and van Norden (2018), the surplus is receipts minus expenditures. I express both as a percentage of nominal GDP in the quarter they refer to.[7]

In September 2017, the staff table was redesigned. Since then it reports the unified federal budget by fiscal year (October to September), already as a percentage of GDP, and a cyclically adjusted surplus in place of the high-employment surplus. For Tealbooks from September 2017 onward, I assign to each quarter the forecast for the fiscal year that contains it, so that both tables deliver forecasts for the same quarterly horizons. Because the two tables measure the budget on different bases, their levels are not directly comparable. I therefore compute revisions only between Greenbooks that use the same table, and in samples that span both tables I add a dummy D_m equal to one for Tealbooks with the redesigned table. The first Tealbook with the new table has no revision and drops out of the sample.

Each announcement is matched to the most recent Greenbook prepared before it. For scheduled meetings this is the Greenbook prepared for that meeting; for unscheduled announcements it is the Greenbook of the preceding meeting, provided it was released at most 60 days earlier. The macroeconomic and the fiscal forecasts come from the same Greenbook.

2.3 The informationally robust instrument

FOMC announcements, scheduled or unscheduled, are indexed by m, and F^{cb}_m denotes the Greenbook forecast matched to announcement m, that is, the most recent Greenbook prepared before it.

In the first step, I regress the $FF4$ surprise on the Greenbook forecasts and on their revisions.

$$FF4_m = \alpha_0 + \sum_{j=-1}^{3} \theta_j F^{cb}_m g_{q+j} + \sum_{j=-1}^{3} \gamma_j F^{cb}_m \pi_{q+j} + \rho F^{cb}_m u_q + \sum_{j=-1}^{2} \vartheta_j' \Delta F^{cb}_m x_{q+j} + MPI_m \quad (1)$$

where g denotes real GDP growth, π GDP-deflator inflation, u the unemployment rate, and $x = (g, \pi, u)'$. The index q denotes the quarter in which Greenbook F^{cb}_m is prepared, and $\Delta F^{cb}_m x_{q+j} = F^{cb}_m x_{q+j} - F^{cb}_{m-1} x_{q+j}$ is the revision relative to the previous Greenbook, F^{cb}_{m-1}. The residual MPI_m is the component of the surprise orthogonal to the Fed's information set, as captured by its forecasts and their revisions.

[7] Nominal GDP is the current vintage from FRED, the same for every Greenbook, so that the ratio moves only with the staff's fiscal forecast.

To test whether the Fed's fiscal forecasts belong to this information set, I add them to equation (1) in the same way as the macroeconomic forecasts, in levels and in revisions:

$$FF4_m = \alpha_0 + \ldots + \sum_{j=-1}^{3} \mu_j' F_m^{cb} f_{q+j} + \sum_{j=-1}^{2} \nu_j' \Delta F_m^{cb} f_{q+j} + \delta D_m + MPI_m^f \tag{2}$$

where the dots stand for the regressors of equation (1), and f is the forecast surplus s, the forecast high-employment surplus h, or both, $f = (s, h)'$, all as a percentage of GDP. If the fiscal forecasts carry information that markets learn from FOMC announcements, or if the Fed's response to them is not anticipated, they will predict the surprises. I test the null hypothesis $\mu_j = \nu_j = 0$ for all j with the partial F-test between equations (2) and (1), estimated on the same sample, and with a Wald test based on heteroskedasticity-robust standard errors. Under the null, MPI_m^f and MPI_m coincide up to sampling error.

In the second step, I aggregate the residuals to the monthly frequency by summing them within each calendar month t:

$$\overline{MPI}_t = \sum_{m \in \mathcal{M}_t} MPI_m, \qquad \overline{MPI}_t = 0 \text{ if } \mathcal{M}_t = \varnothing \tag{3}$$

where $\mathcal{M}_t$ is the set of announcements in month t. In the third step, to account for the slow absorption of information by market participants, I regress the monthly series on twelve of its own lags:

$$\overline{MPI}_t = \phi_0 + \sum_{j=1}^{12} \phi_j \overline{MPI}_{t-j} + MPI_t \tag{4}$$

I estimate equation (4) only on months in which the dependent variable is nonzero. The monthly monetary policy instrument MPI_t is the residual from this regression, and is set to zero in months without announcements. The same two steps applied to MPI_m^f give the instrument with fiscal information, MPI_t^f.[8]

[8] To construct the Fed information instrument, I estimate a more parsimonious version of equation (1),

$$FF4_m = \beta_0 + \sum_{j=-1}^{0} \left(\delta_j F_m^{cb} g_{q+j} + \eta_j F_m^{cb} \pi_{q+j} \right) + \kappa F_m^{cb} u_q + \sum_{j=0}^{1} \lambda_j' \Delta F_m^{cb} x_{q+j} + e_m,$$

and take the fitted values, including the constant: $INFO_m = \widehat{FF4}_m$. These are aggregated to the monthly frequency and purged of their own autoregressive component in the same way as the monetary policy instrument; the resulting residual is the information instrument $INFO_t$. The version with fiscal information, $INFO_t^f$, adds the surplus and the high-employment surplus with the same horizons, $\sum_{j=-1}^{0} \omega_j' F_m^{cb} f_{q+j} + \sum_{j=0}^{1} \psi_j' \Delta F_m^{cb} f_{q+j}$ with $f = (s, h)'$, and the dummy D_m.

I estimate the regression on three samples. The first, 1990-2008, ends with the November 2008 meeting, before the federal funds rate reached the zero lower bound. The second, 1990-July 2017, ends with the last Tealbook that uses the original fiscal table. The third, 1990-2020, includes the redesigned table and the COVID-19 period. In each sample, the regressions with and without the fiscal forecasts are estimated on the same announcements, namely those for which all regressors are available.

Figure 1 shows the monthly instruments MPI_t and $INFO_t$ constructed in this way, with the information set of Miranda-Agrippino and Ricco (2021), from January 1991 to December 2020.

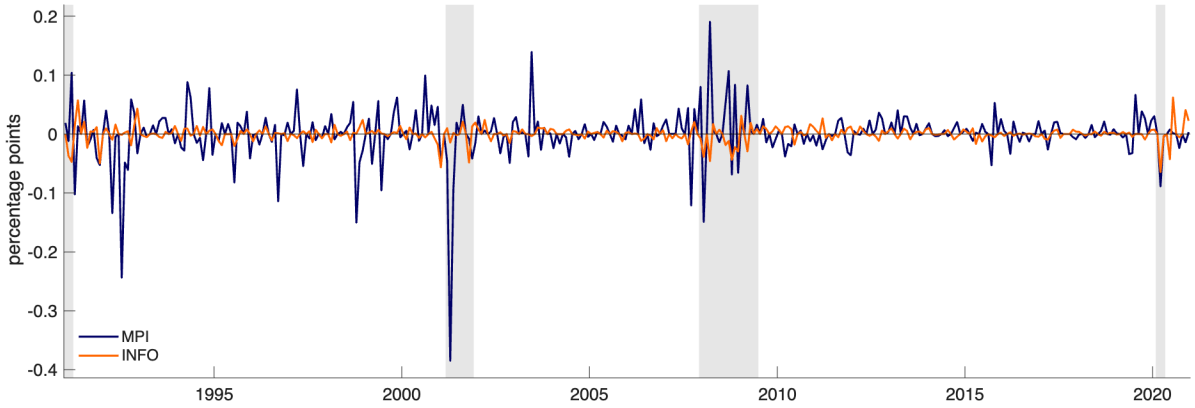


Figure 1: Monetary policy and Fed's information IVs

Notes: Monthly monetary policy instrument MPI_t (dark blue) and information instrument $INFO_t$ (orange), constructed from the FF4 surprises of Jarociński and Karadi (2020) with the Greenbook information set of Miranda-Agrippino and Ricco (2021), 1991:01–2020:12, in percentage points. Shaded areas are NBER recessions.

3 Results

Table 1 reports the estimates of equations (1) and (2) on the three samples. Columns (1), (5) and (9) correspond to the specification of Miranda-Agrippino and Ricco (2021), and the remaining columns add the forecast surplus (S), the forecast high-employment surplus (H), or both. Since each fiscal variable enters with nine coefficients, the table summarises them by the sums of the level and of the revision coefficients, while the tests are on all coefficients jointly.

The fiscal forecasts do not predict the surprises. None of the sums of coefficients

Table 1: Fiscal forecasts in the information set of Miranda-Agrippino and Ricco (2021)

	1990–2008:11				1990–2017:07				1990–2020			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Fiscal forecasts added	none	S	H	S, H	none	S	H	S, H	none	S	H	S, H
Levels, S		−0.49		−0.67		−0.40		−0.58		−0.34		−0.44
		(0.60)		(1.04)		(0.40)		(0.61)		(0.39)		(0.53)
Revisions, S		−0.21		1.90		0.77		−0.85		0.69		−0.43
		(2.08)		(3.38)		(1.30)		(2.26)		(0.94)		(2.01)
Levels, H			−0.64	0.94			−0.43	0.33			−0.36	0.12
			(1.00)	(1.88)			(0.56)	(0.92)			(0.48)	(0.72)
Revisions, H			0.08	−1.46			1.37	1.83			1.33	1.94
			(1.99)	(3.23)			(1.26)	(2.13)			(1.01)	(1.96)
Announcements	175	175	175	175	246	246	246	246	276	276	276	276
R^2	0.213	0.237	0.240	0.271	0.198	0.220	0.223	0.234	0.182	0.206	0.210	0.219
Adjusted R^2	0.093	0.065	0.068	0.047	0.115	0.103	0.106	0.080	0.104	0.097	0.102	0.079
Partial F	—	0.50	0.55	0.59	—	0.68	0.75	0.54	—	0.79	0.95	0.61
p-value	—	0.874	0.836	0.903	—	0.726	0.663	0.937	—	0.628	0.485	0.887
Robust F	—	0.34	0.35	0.64	—	0.57	0.64	0.66	—	0.57	0.84	0.99
p-value	—	0.959	0.958	0.859	—	0.823	0.765	0.844	—	0.825	0.580	0.468
Corr. with baseline MPI_m	—	0.985	0.983	0.962	—	0.986	0.985	0.977	—	0.986	0.983	0.977
Corr. with baseline MPI_t	—	0.977	0.975	0.945	—	0.983	0.982	0.971	—	0.983	0.980	0.972

Notes: Columns (1), (5) and (9) report estimates of equation (1), the remaining columns estimates of equation (2), with $f = s$ in columns marked S, $f = h$ in columns marked H, and $f = (s, h)'$ in columns marked S, H. The dependent variable is the FF4 surprise around FOMC announcements from Jarociński and Karadi (2020). For each fiscal variable, the table reports the sum of the level coefficients μ_j and the sum of the revision coefficients ν_j, in basis points per percentage point of GDP, with heteroskedasticity-robust (HC1) standard errors in parentheses. Partial F is the classical test of $\mu_j = \nu_j = 0$ for all j against equation (1) on the same sample, and robust F the corresponding Wald statistic with HC1 covariance, divided by the number of restrictions. The last two rows report the correlation of MPI_m^f with MPI_m and of the corresponding monthly instruments MPI_t^f and MPI_t. In samples that span the redesign of the Tealbook fiscal table in September 2017, every column includes the dummy D_m. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

is statistically different from zero, and the joint tests are far from rejecting the null that all fiscal coefficients are zero. The partial F-statistics range from 0.51 to 0.94, with p-values between 0.49 and 0.94, and the heteroskedasticity-robust Wald tests give p-values between 0.47 and 0.95. The result holds before the zero lower bound, up to the redesign of the Tealbook fiscal table in 2017, and on the full sample, which includes the redesigned table and the COVID-19 period. Adding nine or eighteen regressors raises the R^2 by 0.02 to 0.06, no more than one would expect from regressors with no explanatory power, and the adjusted R^2 falls in every column.

As a consequence, the instrument is essentially unchanged. The residual of equa-

tion (2), MPI_m^f, has a correlation of at least 0.96 with the baseline residual MPI_m, and after aggregation and the AR(12) step the monthly instruments MPI_t^f and MPI_t have a correlation of at least 0.94. Including the Fed's fiscal forecasts in the information set therefore leaves the informationally robust instrument of Miranda-Agrippino and Ricco (2021) unaffected.

4 Implications for the transmission of monetary policy

The tests in Section 3 show that the fiscal forecasts do not predict the surprises. Even a small residual correlation, however, could matter for the estimated responses (Miranda-Agrippino and Ricco, 2023). As a final check, I therefore use the instruments with and without the fiscal forecasts to identify a monetary policy shock in the VAR of Miranda-Agrippino and Ricco (2021), and compare the results.[9]

The VAR includes industrial production, the consumer price index, the unemployment rate, the 1-year and the 10-year Treasury rates, the S&P 500 index, the real effective exchange rate and the excess bond premium of Gilchrist and Zakrajšek (2012), at the monthly frequency from 1979 to 2019, with twelve lags. The VAR is estimated with Bayesian methods, using a Normal-Inverse Wishart prior. The monetary policy shock is identified with each instrument in turn, and normalised to raise the 1-year rate by 100 basis points on impact.

Table 2 reports the strength of the two instruments. The monetary policy instrument built with the information set of Miranda-Agrippino and Ricco (2021) has a first-stage F-statistic of 12.2 at the posterior mode, and the one built with the fiscal forecasts has an F-statistic of 10.7. The 90% posterior bands of the two statistics largely overlap, and the reliability of the two instruments, that is, the share of the variance of the instrument explained by the shock, is nearly identical.[10]

Figure 2 shows the responses to the monetary policy shock identified with the two instruments. The responses have the same shape for every variable. A monetary tightening lowers industrial production and prices, raises unemployment, depresses stock prices, appreciates the dollar and widens the excess bond premium. The two

[9]The specification, including the variables, the lag length and the prior, is the one used to produce Figure 6 in Miranda-Agrippino and Ricco (2021). The only difference is the sample, which here runs from 1979 to 2019.

[10]Extending the sample to 2020 strengthens both instruments, and the comparison between them is unchanged (Table 4 in the Appendix).

Table 2: First-stage diagnostics in the proxy VAR

	F-statistic		Reliability	
Instrument	Mode	90% band	Mode	90% band
MPI_t, MAR information set	12.19	[5.42, 14.87]	0.22	[0.16, 0.24]
MPI_t^f, with fiscal forecasts	10.66	[4.59, 13.21]	0.23	[0.17, 0.24]

Notes: First-stage *F*-statistic and reliability of the monetary policy instrument in the proxy VAR, at the posterior mode and 90% posterior bands. Reliability is the share of the variance of the instrument explained by the identified shock (Mertens and Ravn, 2013).

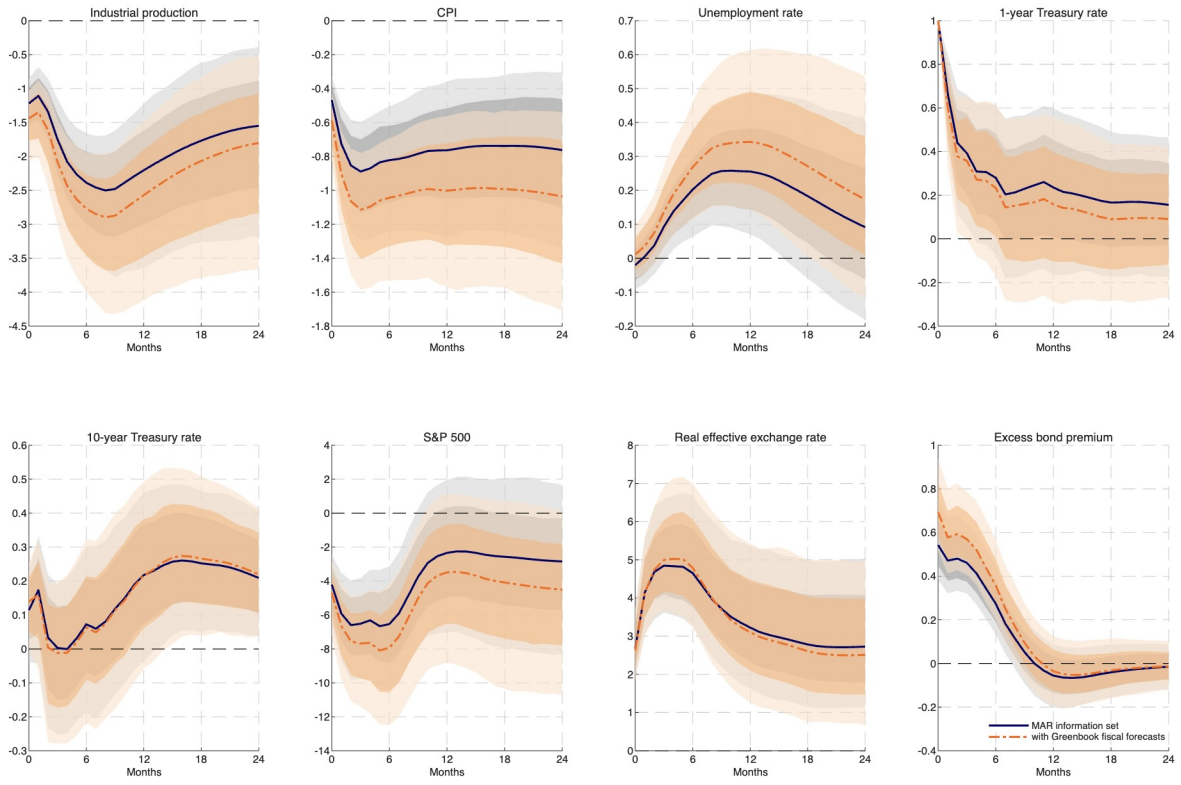


Figure 2: Responses to a monetary policy shock, with and without fiscal forecasts

Note: Responses to a monetary policy shock that raises the 1-year Treasury rate by 100 basis points on impact, identified with the instrument built with the information set of Miranda-Agrippino and Ricco (2021) (solid dark blue line, grey bands) and with the instrument that also includes the Greenbook fiscal forecasts (dash-dotted orange line, orange bands). Lines are point estimates, shaded areas are 68% and 90% posterior bands.

sets of responses differ mainly in size. With the fiscal forecasts in the information set, the contraction in industrial production and prices, the rise in unemployment and the decline in stock prices are somewhat larger, and the response of the 1-year rate is slightly less persistent, while the responses of the 10-year rate and of the exchange rate are nearly identical. The differences are small relative to the posterior uncertainty. At every horizon, the gap between the two point estimates is smaller than the width of the 68% band of the baseline, and the two sets of 90% bands overlap almost entirely. The main exceptions are the price level at longer horizons and the excess bond premium on impact, where the response with the fiscal forecasts lies just outside the 68% band of the baseline. Including the fiscal forecasts in the information set therefore leaves the transmission of monetary policy shocks essentially unchanged.

5 Conclusion

Miranda-Agrippino and Ricco (2021) build an informationally robust instrument for monetary policy shocks by purging high-frequency surprises of the Fed's macroeconomic forecasts and of their own past. In their footnote 16, they point out that the instrument would remain contaminated if the Fed also conditioned its policy on variables outside this information set, such as fiscal variables. This note tests that possibility, using the Greenbook fiscal forecasts collected by Croushore and van Norden (2018) and extended here to 2020.

The fiscal forecasts do not predict the surprises of Jarociński and Karadi (2020), whether the surplus, the high-employment surplus, or both are added to the information set, and whatever the sample considered. Including them leaves the instrument essentially unchanged, and in a proxy VAR the instruments with and without the fiscal forecasts have similar strength and deliver responses to a monetary policy shock that differ little. The result does not imply that the Fed ignores the fiscal outlook, which Croushore and van Norden (2018) show helps explain the FOMC's policy decisions. It is consistent instead with markets anticipating this part of the Fed's behaviour, or with the staff's fiscal forecasts revealing little that markets did not already know about the budget. In either case, the fiscal outlook leaves no trace in the surprises, and for applied work the informationally robust instrument of Miranda-Agrippino and Ricco (2021) needs no fiscal correction.

The analysis has some limitations, which also point to extensions. It considers two fiscal variables, the surplus and the high-employment surplus, and does not exploit

other items in the staff's fiscal projections, such as federal debt, which is reported from 2017. It relies on a single measure of the surprises, and the redesign of the Tealbook fiscal table in 2017 makes the most recent years harder to compare with the earlier ones. Finally, the extended Greenbook fiscal forecasts may be useful beyond this note, for instance to study how the Fed's view of fiscal policy evolved around the zero lower bound and the COVID-19 pandemic.

References

Bauer, M. D. and Swanson, E. T. (2023). An alternative explanation for the "Fed information effect". *American Economic Review*, 113(3):664–700.

Croushore, D. and van Norden, S. (2018). Fiscal forecasts at the FOMC: Evidence from the Greenbooks. *Review of Economics and Statistics*, 100(5):933–945.

Croushore, D. and van Norden, S. (2019). Fiscal surprises at the FOMC. *International Journal of Forecasting*, 35(4):1583–1595.

Gertler, M. and Karadi, P. (2015). Monetary policy surprises, credit costs, and economic activity. *American Economic Journal: Macroeconomics*, 7(1):44–76.

Gilchrist, S. and Zakrajšek, E. (2012). Credit spreads and business cycle fluctuations. *American Economic Review*, 102(4):1692–1720.

Gürkaynak, R. S., Sack, B., and Swanson, E. T. (2005). Do actions speak louder than words? The response of asset prices to monetary policy actions and statements. *International Journal of Central Banking*, 1(1):55–93.

Jarociński, M. and Karadi, P. (2020). Deconstructing monetary policy surprises—the role of information shocks. *American Economic Journal: Macroeconomics*, 12(2):1–43.

Kuttner, K. N. (2001). Monetary policy surprises and interest rates: Evidence from the Fed funds futures market. *Journal of Monetary Economics*, 47(3):523–544.

Mertens, K. and Ravn, M. O. (2013). The dynamic effects of personal and corporate income tax changes in the United States. *American Economic Review*, 103(4):1212–1247.

Miranda-Agrippino, S. and Ricco, G. (2021). The transmission of monetary policy shocks. *American Economic Journal: Macroeconomics*, 13(3):74–107.

Miranda-Agrippino, S. and Ricco, G. (2023). Identification with external instruments in structural VARs. *Journal of Monetary Economics*, 135:1–19.

Nakamura, E. and Steinsson, J. (2018). High-frequency identification of monetary non-neutrality: The information effect. *Quarterly Journal of Economics*, 133(3):1283–1330.

Romer, C. D. and Romer, D. H. (2000). Federal Reserve information and the behavior of interest rates. *American Economic Review*, 90(3):429–457.

Romer, C. D. and Romer, D. H. (2004). A new measure of monetary shocks: Derivation and implications. *American Economic Review*, 94(4):1055–1084.

A Data

Table 3: Variables in the VAR

Variable	Code	Source	Transformation	Random-walk prior
Industrial production	`INDPRO`	FRED	$100 \times \log$	Yes
Consumer price index	`CPIAUCSL`	FRED	$100 \times \log$	Yes
Unemployment rate	`UNRATE`	FRED	Level	No
1-year Treasury rate	`DGS1`	FRED	Level	No
10-year Treasury rate	`GS10`	FRED	Level	No
S&P 500	—	FRED	$100 \times \log$	Yes
Real effective exchange rate	—	BIS	$100 \times \log$	Yes
Excess bond premium	`EBP`	FRB	Level	No

Notes: Random-walk prior indicates the variables for which the prior mean of the first own lag is one; for the others it is zero. FRED is the Federal Reserve Economic Data of the Federal Reserve Bank of St. Louis, BIS the Bank for International Settlements, and FRB the Board of Governors of the Federal Reserve System. The excess bond premium is that of Gilchrist and Zakrajšek (2012).

B F-statistic on the sample 1979–2020

Table 4: First-stage diagnostics in the proxy VAR, 1979–2020

	F-statistic		Reliability	
Instrument	Mode	90% band	Mode	90% band
MPI_t, MAR information set	17.80	[8.65, 19.46]	0.21	[0.14, 0.22]
MPI_t^f, with fiscal forecasts	14.62	[6.84, 16.33]	0.21	[0.14, 0.22]

Notes: First-stage F-statistic and reliability of the monetary policy instrument in the proxy VAR of Table 2, estimated on the sample 1979–2020, at the posterior mode and 90% posterior bands. Reliability is the share of the variance of the instrument explained by the identified shock (Mertens and Ravn, 2013).

C First-stage diagnostics of the information instrument

Table 5: First-stage diagnostics of the information instrument in the proxy VAR

	F-statistic		Reliability	
Instrument	Mode	90% band	Mode	90% band
Sample 1979–2019				
$INFO_t$, MAR information set	5.74	[0.94, 10.57]	0.16	[0.12, 0.20]
$INFO_t^f$, with fiscal forecasts	6.95	[1.45, 11.65]	0.15	[0.10, 0.19]
Sample 1979–2020				
$INFO_t$, MAR information set	7.43	[1.06, 14.23]	0.25	[0.20, 0.30]
$INFO_t^f$, with fiscal forecasts	14.34	[5.04, 19.30]	0.29	[0.23, 0.32]

Notes: First-stage *F*-statistic and reliability of the information instrument in the proxy VAR, at the posterior mode and 90% posterior bands, on the samples 1979–2019 and 1979–2020. The shock identified by the information instrument is normalised to raise the 1-year Treasury rate by 100 basis points on impact. Reliability is the share of the variance of the instrument explained by the identified shock (Mertens and Ravn, 2013).